# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The 6th term of an AP is $20$ and the 10th term is $32$. The 16th term is: 1

**(a)** $50$ **(b)** $47$ **(c)** $56$ **(d)** $53$

**2.** The sum of the first $n$ terms of a sequence is $S_n = 2n^2 + 3n$. Its 9th term is: 1

**(a)** $37$ **(b)** $39$ **(c)** $41$ **(d)** $35$

**3.** The sum of all integers from 1 to 100 that are divisible by 6 is: 1

**(a)** $810$ **(b)** $826$ **(c)** $816$ **(d)** $822$

**4.** The sum of the first 12 terms of an AP is $510$ and its first term is $4$. The common difference is: 1

**(a)** $7$ **(b)** $6$ **(c)** $8$ **(d)** $9$

**5.** Three numbers in AP have sum $30$ and product $840$ (with $d > 0$). The largest number is: 1

**(a)** $15$ **(b)** $14$ **(c)** $10$ **(d)** $18$

**6.** A town's population is 10000 and grows by 10% each year. After 2 years it is: 1

**(a)** $12100$ **(b)** $12200$ **(c)** $12000$ **(d)** $12300$

**7.** The first term of a GP is $3$ and the 8th term is $384$ (with $r > 0$). The common ratio is: 1

**(a)** $1$ **(b)** $2$ **(c)** $4$ **(d)** $3$

**8.** The $n$th term of an AP is $t_n = 3n + 2$. The sum of its first 16 terms is: 1

**(a)** $443$ **(b)** $437$ **(c)** $456$ **(d)** $440$

**9.** The 5th term of an AP is $44$ and the 12th term is $16$. The 17th term is: 1

**(a)** $-4$ **(b)** $0$ **(c)** $-12$ **(d)** $-8$

**10.** The sum of the first $n$ terms of a sequence is $S_n = 3n^2 + 1n$. Its 12th term is: 1

**(a)** $73$ **(b)** $67$ **(c)** $76$ **(d)** $70$

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** The sum of the first 15 terms of an AP is $255$ and the first term is $3$. Find $d$ and the 15th term. 2

**12.** Which term of the AP $5, 11, 17, \ldots$ is the first term to exceed $200$? 2

**13.** If $S_n = 3n^2 + 4n$ is the sum of the first $n$ terms of a sequence, find its first term and its 10th term. 2

**14.** The sum of the first 12 terms of an AP is $336$ and the first term is $50$. Find $d$ and the 12th term. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** The sum of the 3rd and 9th terms of an AP is $40$ and its 6th term is $20$. Find the AP's first term and the sum of its first 20 terms. **3**

**16.** Find the sum of all natural numbers between 10 and 200 that leave remainder 2 when divided by 5. **3**

**17.** Three positive numbers are in GP. Their product is $216$ and the sum of the first and third is $15$. Find the numbers. **3**

**OR**

The sum of the 2nd and 10th terms of an AP is $64$ and its 5th term is $28$. Find the AP's first term and the sum of its first 14 terms. **3**

**18.** Find the sum of all natural numbers between 20 and 300 that leave remainder 3 when divided by 7. **3**

**OR**

Three positive numbers are in GP. Their product is $1728$ and the sum of the first and third is $40$. Find the numbers. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A contractor pays a penalty for delay: ₹200 on day 1, ₹250 on day 2, ₹300 on day 3 and so on. If the work is 30 days late, find the penalty on the last day and the total penalty. **5**

**OR**

A ball is dropped from 81 m and rebounds to 2/3 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. **5**

**20.** A contractor pays a penalty for delay: ₹100 on day 1, ₹125 on day 2, ₹150 on day 3 and so on. If the work is 24 days late, find the penalty on the last day and the total penalty. **5**

**OR**

A ball is dropped from 64 m and rebounds to 3/4 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. **5**

## END OF QUESTION PAPER — Total: 40 Marks

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